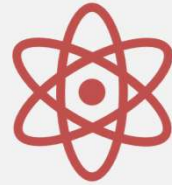
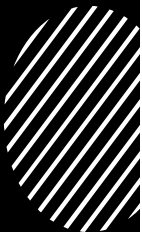




SpaceX Launch Success Prediction



End-to-end Data
Science Project



Author: Juan
Angel Iglesias



Goal: Predict
Falcon 9 landing
success

Executive Summary



Built full pipeline: Data Collection
→ Wrangling → EDA → SQL → ML



Analyzed launch site, payload,
orbit and booster version



Dashboard enabled interactive
exploration



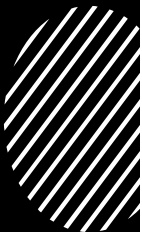
Best ML model achieved ~85%
accuracy



Key insight: payload and orbit
strongly influence success



Data Sources



SpaceX REST API
→ structured
launch data

Wikipedia Web
Scraping →
historical missions

Dataset includes:

- Payload mass
(kg)

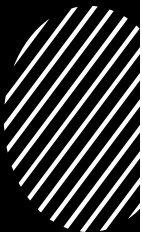
- Launch site

- Orbit type

- Booster version



Data Wrangling Process



Removed missing values and inconsistent entries

Converted categorical variables to numerical (encoding)

Feature engineering:

- Created binary target (success/failure)

- Normalized payload ranges

Prepared dataset for ML models

EDA Methodology



Used Pandas, Matplotlib and Seaborn



Analyzed relationships between variables:



- Payload vs Success



- Orbit vs Success



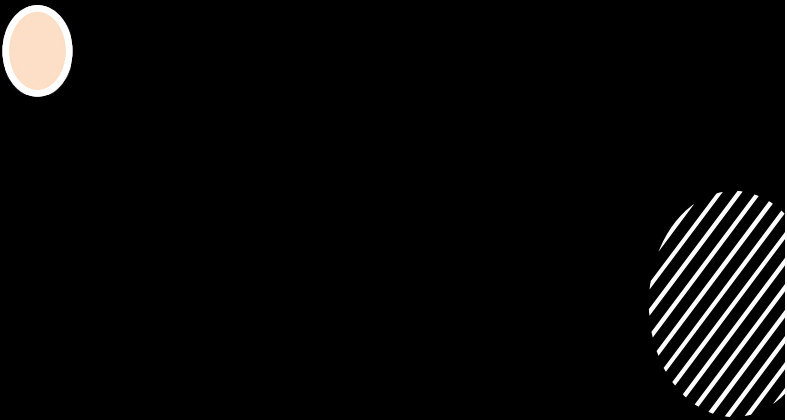
- Launch Site vs Success



Identified trends and correlations



EDA Results



Higher payload generally
reduces success probability

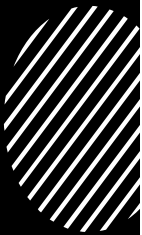
LEO orbit shows highest
success rate

KSC LC-39A has strongest
performance

Clear clustering observed in
payload-success scatter



SQL Analysis



Performed queries on dataset using SQL



Key queries:



- Count launches per site



- Average payload mass



- Success rate per orbit



Enabled structured insights beyond visualization

SQL Results



KSC LC-39A → highest
success rate



CCAFS LC-40 → most
launches

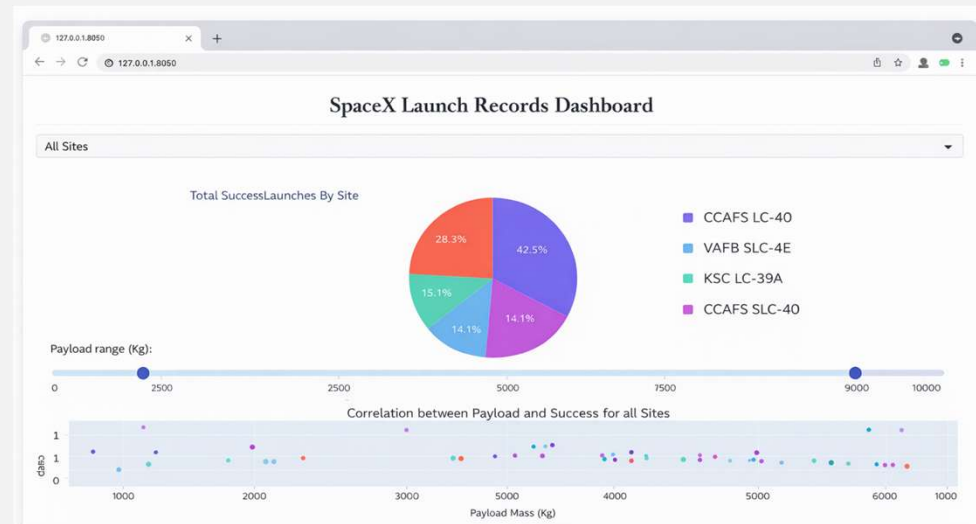


LEO → most frequent and
successful orbit



Payload distribution varies
significantly by mission

Interactive Dashboard (Plotly Dash)



- Dropdown → filter by launch site
- Slider → filter payload range
- Pie chart → success distribution
- Scatter plot → payload vs success

Dashboard Insights

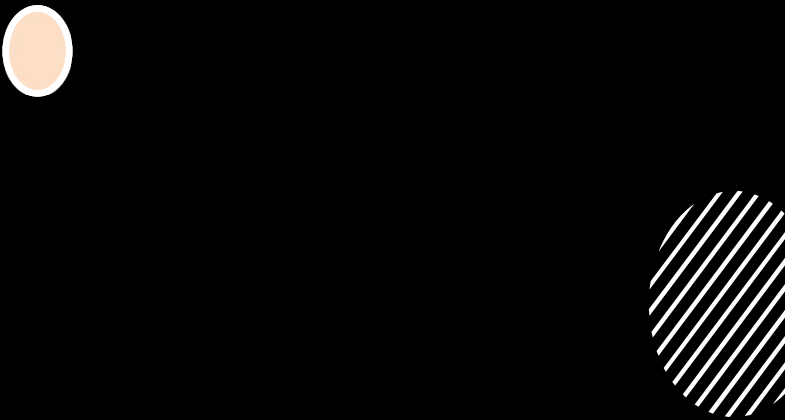
Here are the answers in English based on the dashboard analysis described in the lab:

1. Which site has the largest successful launches?
CCAFS LC-40 has the highest number of successful launches.
2. Which site has the highest launch success rate?
KSC LC-39A shows the highest success rate (highest proportion of successful launches vs failures).
3. Which payload range(s) has the highest launch success rate?
Payloads between 4000 kg and 6000 kg tend to have the highest success rate.
4. Which payload range(s) has the lowest launch success rate?
Payloads below 2000 kg generally show lower success rates.
5. Which F9 Booster version has the highest launch success rate?
The Falcon 9 FT (Full Thrust) and newer versions (especially Block 5) have the highest success rates.

-
- CCAFS LC-40 → most successful launches
 - KSC LC-39A → highest success rate
 - Best payload range: 4000–6000 kg
 - Low success for payload <2000 kg
 - Best boosters: Falcon 9 FT / Block 5



Machine Learning Methodology



Split dataset into
Train/Test sets

Applied feature scaling

Used GridSearchCV for
hyperparameter tuning

Evaluated multiple
classification models

Model Performance Comparison



Logistic Regression → ~80% accuracy



SVM → ~85% accuracy (best)



Decision Tree → ~78% accuracy



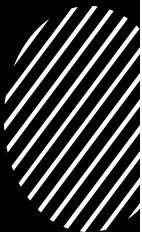
KNN → ~82% accuracy



SVM selected as final model



Model Insights



Most important features:

- Orbit type

- Payload mass

- Launch site

Model generalizes well on test data

No significant overfitting observed

Conclusion



Data-driven approach successfully predicts landing outcome



Operational insights can reduce risk and cost



Approach scalable to future missions

Future Improvements



Include weather
conditions



Include booster reuse
cycles



Deploy real-time
prediction system



Integrate with mission
planning tools